ACTIVE LEARNING

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Active Learning is a machine learning approach that reduces the need for large labeled datasets by selectively querying only the most informative samples. This makes it especially useful for tasks such as **credit card fraud detection**, where class imbalance is significant and labeling requires expert validation.

1 Introduction

Machine learning has become a powerful tool in solving large-scale classification problems. Among these, **credit card fraud detection** presents a critical challenge due to severe class imbalance and constantly changing fraudulent patterns. Traditional supervised learning approaches assume the availability of large, fully-labeled datasets. In practice, however, obtaining labeled fraud data is expensive and time-consuming, because it often requires manual verification by domain experts.

Active Learning (AL) addresses this bottleneck by enabling the model to request labels only for data points that are expected to be the most informative. Instead of passively consuming whatever labeled data is available, the model actively identifies uncertain instances and queries labels for them. This helps reduce annotation cost while maintaining, or even improving, model performance.

In the context of fraud detection, where unlabeled transaction data is abundant but correctly labeled fraud cases are rare, Active Learning becomes particularly suitable. This report presents the theoretical foundations of Active Learning strategies followed by an experimental evaluation on a real-world credit card fraud dataset.

2 Theoretical and Mathematical Background

Active Learning operates on two pools of data:

$$L = \{(x_i, y_i)\} \quad \text{(labeled set)}, \qquad U = \{x_j\} \quad \text{(unlabeled set)}.$$

The goal is to find the most informative point using a query function:

$$x^* = \arg\max_{x \in U} Q(x),$$

where the scoring function $Q(x)$ measures usefulness, uncertainty, or expected error reduction.

A probabilistic classifier returns label likelihoods:

$$p(y|x) = f(x; \theta),$$

where θ are model parameters. A prediction with lower confidence implies **higher uncertainty**, which is desirable in Active Learning selection.

Uncertainty-Based Sampling Strategies (Used in This Work)

Since fraud detection datasets are highly imbalanced, the following uncertainty-based strategies were implemented:

(a) Least Confidence Sampling selects samples where the model is least confident:

$$LC(x) = 1 - \max_c p(y = c|x).$$

(b) Margin-Based Sampling measures closeness between the top two prediction probabilities:

$$M(x) = p_{(1)}(x) - p_{(2)}(x),$$

where $p_{(1)}(x)$ and $p_{(2)}(x)$ are the highest and second-highest class probabilities. Smaller values indicate higher classifier confusion.

(c) Entropy-Based Sampling considers the full probability distribution:

$$H(x) = -\sum_c p(y = c|x) \log p(y = c|x).$$

This method performed best in our implementation due to its ability to capture prediction uncertainty across all classes.

(d) Random Sampling (Baseline) selects points uniformly at random and is used as a reference to measure the benefit of informed querying.

Strategy	Formula / Criterion	Used in Project
Random Sampling	No mathematical score; uniform selection	Yes (Baseline)
Least Confidence	$LC(x) = 1 - \max p(y\|x)$	Yes
Margin-Based Sampling	$M(x) = p_{(1)} - p_{(2)}$ (small margin $\Rightarrow$ uncertain)	Yes
Entropy Sampling	$H(x) = -\sum p \log p$ (highest entropy preferred)	Yes (Best Performer)

Table 1: Comparison of query strategies used in the project.

Comparison Table of Strategies Used

Label Efficiency Concept

A key advantage of Active Learning is reducing labeling cost. If:

$$L_{SL} = \text{labeled samples needed in supervised learning}, \qquad L_{AL} = \text{samples required with Active Learnin}$$

then ideally:

$$L_{AL} \ll L_{SL}.$$

This difference is known as **label efficiency** and represents the primary motivation for using Active Learning in real-world systems like fraud detection.

Supervised Learning vs Active Learning

Aspect	Traditional Supervised Learning	Active Learning
Labeling Requirement	Full dataset must be labeled before training	Only selected (uncertain) samples are labeled
Cost	High annotation effort and time	Reduced labeling cost and time
Learning Style	Passive: model consumes fixed labeled data	Active: model chooses which points to label
Performance Growth	Often gradual, requires more labels	Faster improvement with fewer labels
Best Use Case	Balanced, fully labeled datasets	Imbalanced or expensive-to-label datasets

Table 2: Comparison of supervised learning and Active Learning.

3 Experimental Setup

The experimental phase of this project focused on evaluating the effectiveness of Active Learning strategies in reducing labeling effort while maintaining high model performance for credit card fraud detection. All experiments were executed in Google Colab using Python. The main libraries used were Scikit-Learn, XGBoost, Imbalanced-Learn, Matplotlib, Seaborn and SHAP.

The goal of the setup was not only to train a fraud classifier but also to study how model performance evolves when new labeled samples are selected *intelligently* rather than randomly. This mimics realistic financial systems where unlabeled transaction data is plentiful, but verified fraud labels are scarce and costly to obtain.

Resources

Google Colab Notebook: https://colab.research.google.com/drive/1CQxKz2XpfmefJAZpqkXWG_S ?usp=sharing

Dataset Source (Kaggle): https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Step 1: Data Acquisition and Understanding

The dataset used is the Kaggle Credit Card Fraud Detection dataset, consisting of 284,807 anonymized transactions with 30 numerical features derived from PCA transformations and one binary target label. Only 492 samples correspond to fraudulent transactions, making the problem extremely imbalanced.

Exploratory Data Analysis (EDA) was first performed to understand the data distribution and class imbalance. A class distribution plot clearly confirmed that fraud represents less than 0.2% of all records. In addition, transaction amount distribution was examined separately for fraud and non-fraud classes to observe differences in value ranges and density.

These visualizations motivated the need for both imbalance handling techniques and label-efficient learning strategies.

Step 2: Preprocessing and SMOTE

Since raw features are on different scales, all numerical attributes were standardized using **StandardScaler**. The dataset was then split into training and testing subsets using stratified sampling so that the original fraud ratio was preserved.

Due to the extreme class imbalance, the classifier would otherwise tend to predict only the majority class. To address this, the **SMOTE (Synthetic Minority Oversampling**

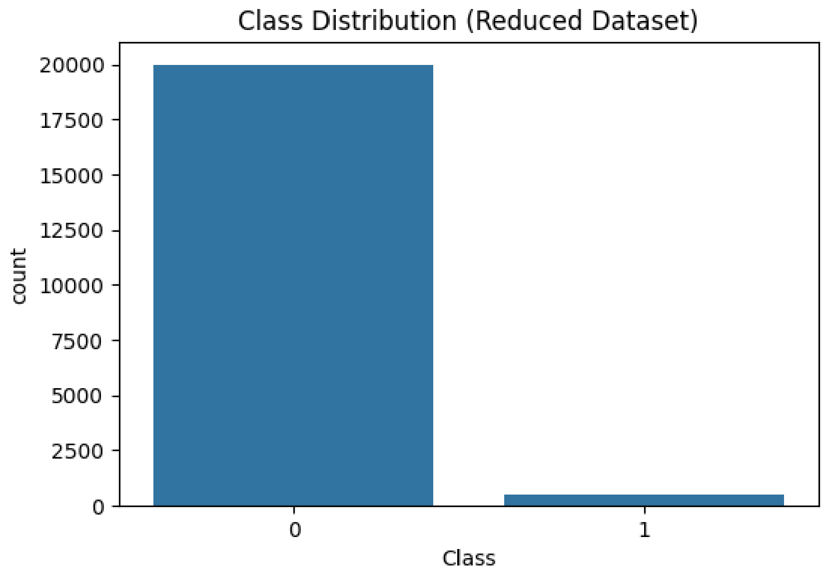

Figure 1: Class distribution highlighting extreme imbalance.

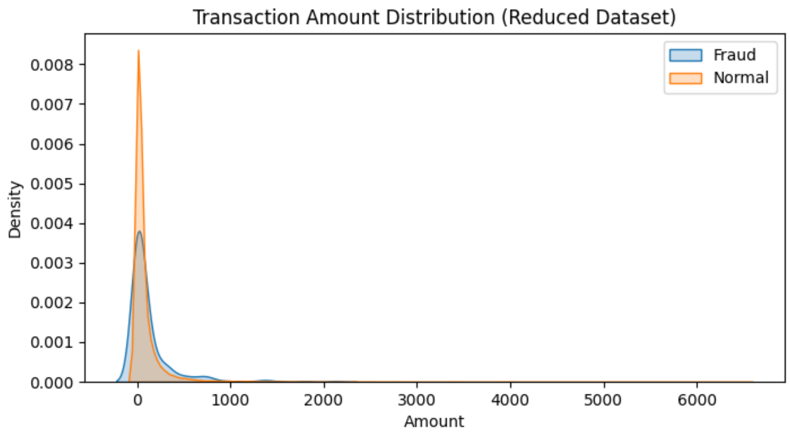

Figure 2: Transaction amount distribution for fraud and non-fraud classes.

Technique) algorithm was applied to only the training set. SMOTE synthetically generates new minority class samples by interpolating between existing fraud points. This increases the representation of the fraud class and helps the classifier learn a more balanced decision boundary.

PCA (Principal Component Analysis) was used to visualize the structure of the data before and after applying SMOTE. Before SMOTE, the fraud points were sparse and located in a narrow region surrounded by a dense cloud of normal transactions. After SMOTE, fraudulent samples became more spread out in the feature space, improving separability and enabling better learning.

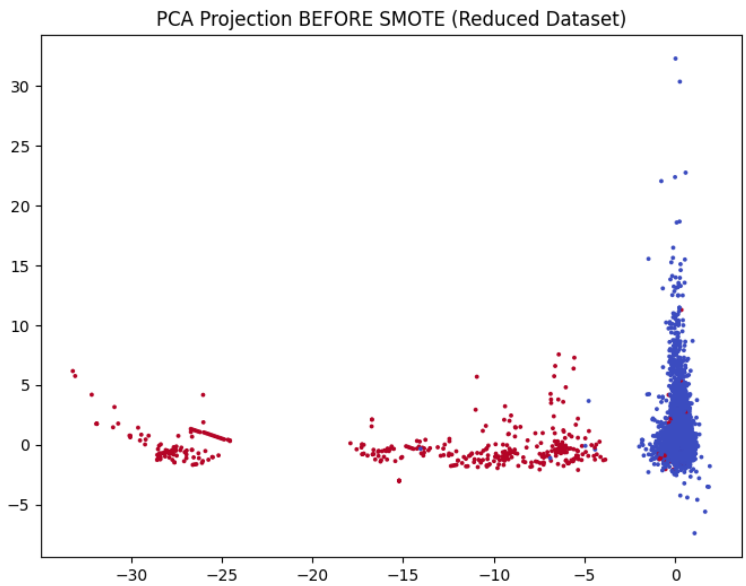

Figure 3: PCA projection before SMOTE (fraud class is very sparse).

Step 3: Baseline Model Selection and Evaluation

Before introducing Active Learning, four supervised models were trained on the preprocessed and balanced training data:

- Logistic Regression
- Random Forest
- XGBoost
- Multilayer Perceptron (MLP) Neural Network

These models were chosen to cover linear, ensemble, boosting and neural approaches. Evaluation was performed on the untouched test set using ROC-AUC and F1-score, which are more appropriate than accuracy in imbalanced settings.

Among all models, **XGBoost** achieved the highest ROC-AUC and a strong F1-score, showing robust performance on both classes. Due to its superior behavior and efficiency, XGBoost was selected as the base classifier for the subsequent Active Learning experiments.

Step 4: Active Learning Experimentation

Active Learning was implemented using a pool-based framework. The initial labeled set consisted of only 800 samples, simulating a scenario where only a small number of

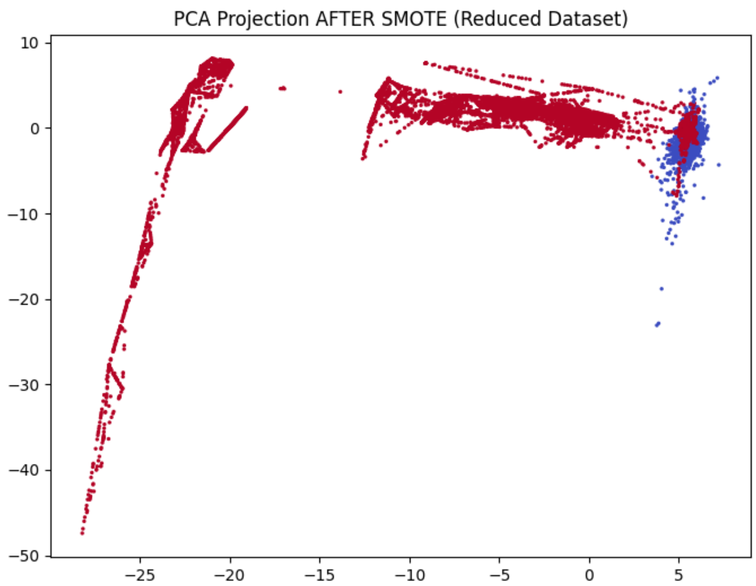

Figure 4: PCA projection after SMOTE (fraud class better represented).

transactions have been manually inspected. The remaining training samples formed the unlabeled pool.

In each Active Learning iteration:

1. The XGBoost model was trained on the current labeled set.
2. Depending on the strategy (Random, Margin, or Entropy), the model selected 250 unlabeled points that it considered most informative.
3. The true labels for these selected points were then revealed (simulated using the dataset).
4. The labeled set was updated and the model was retrained.

This process was repeated for seven iterations, gradually increasing the number of labeled samples while monitoring performance. For each strategy, ROC-AUC and F1-score were recorded after every iteration to obtain learning curves.

The curves clearly show that **entropy-based sampling** achieves faster improvement and reaches higher performance with fewer labeled instances compared to random sampling. Margin-based sampling generally lies between random and entropy, providing moderate label efficiency.

Step 5: Explainability and Feature Importance

Once the best-performing Active Learning strategy had been identified, the final XGBoost model trained with entropy-based sampling was analyzed using **SHAP (SHapley Ad-**

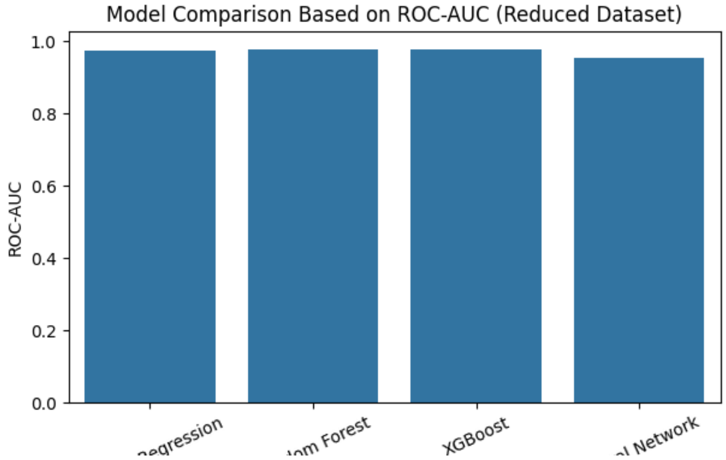

Figure 5: Baseline model comparison using ROC-AUC. XGBoost performs best.

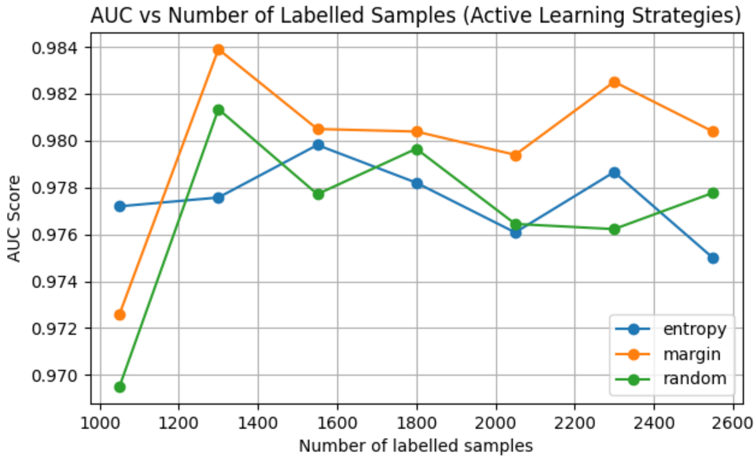

Figure 6: AUC progression across Active Learning iterations for different strategies.

ditive exPlanations). SHAP assigns contribution scores to each feature for individual predictions and provides a global ranking of important features.

The SHAP summary plot revealed which transformed features have the strongest impact on predicting fraud, enhancing trust and interpretability in a high-stakes domain such as financial risk analysis.

Key Observations and Highlights

- Active Learning significantly reduced the number of labeled samples required to reach high fraud detection performance.
- Entropy-based sampling consistently outperformed random and margin-based sampling in terms of label efficiency and final accuracy.
- SMOTE balancing combined with standardized features improved the classifier's ability to learn minority patterns.

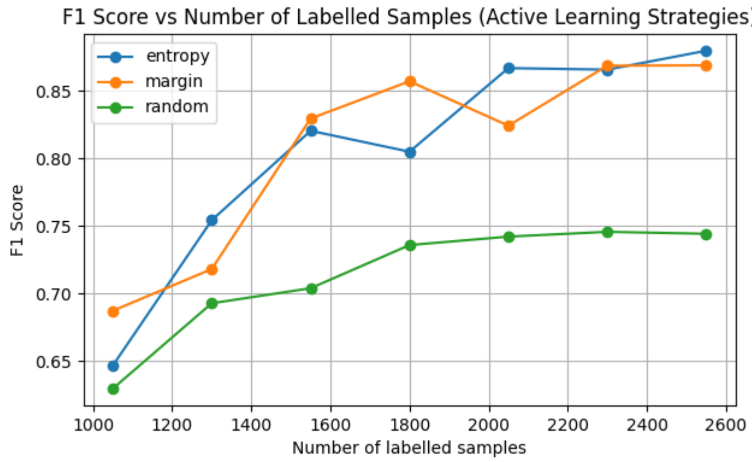

Figure 7: F1-score progression as more labeled samples are added.

- XGBoost served as a strong and stable base model throughout both the baseline and Active Learning phases.
- SHAP-based analysis confirmed that the model relied on meaningful and consistent feature patterns, increasing the reliability of the system.

4 Conclusion

The experimental results show that Active Learning, particularly entropy-based uncertainty sampling, can achieve strong fraud detection performance with significantly fewer labeled samples compared to traditional supervised training. This makes Active Learning a practical and cost-effective approach for real-world fraud detection systems where labeled data is scarce and expensive.

5 Future Work

Future extensions of this work may include:

- Exploring deep learning models within the Active Learning framework.
- Investigating cost-sensitive and risk-aware sampling strategies.
- Extending the method to online or streaming data where fraud patterns evolve over time.

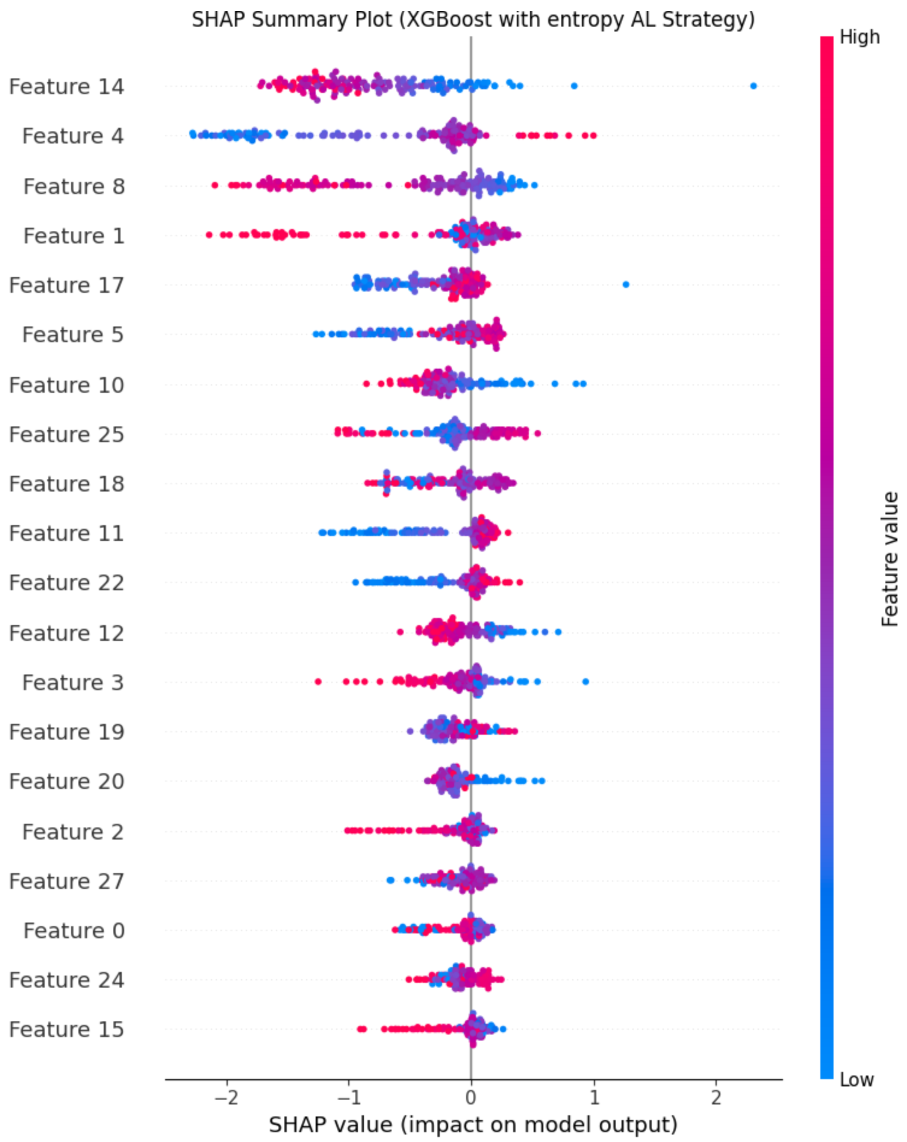

Figure 8: SHAP summary plot showing the most influential features for fraud detection.